SatNOGS Audio to Operational Telemetry

BEACON DECODER

GNU Radio | G3RUH AX.25 | CRC-valid 263-byte frames

5 / 5

observations passed

22

valid local frames

20

byte-exact matches

Final presentation | 31 Aug 2026

Objective and Acceptance Criteria

PROJECT OBJECTIVE

- Decode the beacon from SatNOGS OGG audio.
- Verify locally recovered frames independently.
- Feed traceable telemetry into an operational dashboard.

STRICT PASS CRITERION

- At least one CRC-valid local KOYO frame.
- Expected 263-byte frame, AX.25 path, and type.
- At least one byte-exact match for the same observation.

RESULT

Success is reproducible and measurable, not based on visual similarity.

What Each Stage Means

SIGNAL SIDE

- Beacon: periodic spacecraft status message.
- OGG: audio recorded by a SatNOGS ground station.
- Demodulation: convert 9600-baud FSK audio back to bits.

DATA SIDE

- AX.25/HDLC: frame boundaries, addressing, and CRC.
- Telemetry: interpret payload bytes as engineering values.
- Dashboard: present evidence without inventing data.

RESULT

Audio decoding and telemetry visualization are separate responsibilities.

End-to-End Architecture

SatNOGS OGG
recording + metadata

→

PCM WAV
mono / 48 kHz / PCM16

→

GNU Radio
FSK / 9600 baud

→

AX.25 + KISS
G3RUH / HDLC / CRC

→

KOYO Decoder
263-byte filter

→

Influx + Grafana
evidence + telemetry

INDEPENDENT CONTROL

SatNOGS demodulated frames are used only for byte-exact comparison. They are not local decoder input.

RESULT

OGG -> WAV -> GNU Radio -> AX.25/KISS -> decoder -> dashboard

Stage 1 - SatNOGS Observation Acquisition

DEMO OBSERVATION

- Observation 14909703 from MAUSyagi-AK.
- Start: 2026-08-30 18:43:47 UTC.
- Duration 00:09:09.61; OGG size 10,431,019 bytes.

TRACEABILITY AND SCOPE

- Observation ID, station, and UTC follow every result.
- Same-observation demodulated frames are controls.
- Latest means newest upload, not continuous live RF.

RESULT **Observation 14909703 acquired with complete source metadata.**

Stage 2 - Audio Preparation

SOURCE AUDIO

- Compressed SatNOGS Vorbis OGG.
- 10.4 MB for the demonstration observation.
- Converted automatically with FFmpeg.

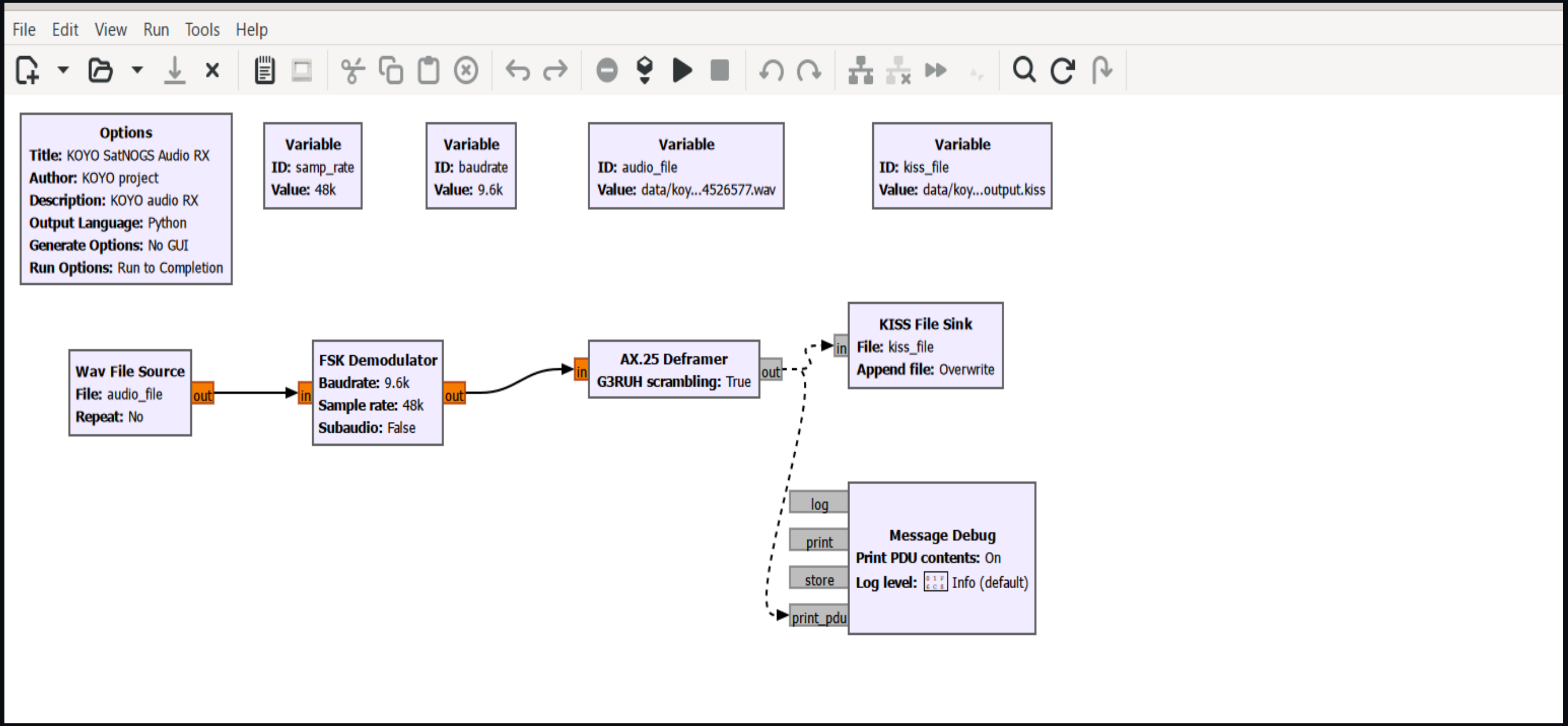
GNU RADIO INPUT

- Mono, 48 kHz, signed 16-bit PCM WAV.
- 52.8 MB uncompressed output.
- Five samples per 9600-baud symbol.

RESULT

A deterministic 48 kHz PCM input is created for GNU Radio.

Stage 3 - GNU Radio Demodulation



PARAMETERS

FSK: 9600 baud
Input: 48 kHz
Deviation: 3 kHz
Clock BW: 0.15

DEFRAMER

NRZI
G3RUH descramble
HDLC + CRC
KISS output

RESULT

Six KISS frames were captured from the demo OGG-derived WAV.

Stage 4 - AX.25 and KOYO Frame Validation

ACCEPTANCE GATES

- CRC accepted by the AX.25 deframer.
- Exactly 263 bytes with KOYOSC -> GS-H20 path.
- Expected control 03 and PID F0 frame type.

DEMO FRAME FUNNEL

- Six KISS frames captured.
- Two frames passed all KOYO checks.
- One frame matched SatNOGS control byte for byte.

RESULT

6 captured -> 2 valid KOYO -> 1 byte-exact control match.

Stage 5 - Multi-Station Audio Validation

OBSERVATION	STATION	CONTROL	LOCAL	EXACT	RECOVERY	RESULT
14526577	MAUSyagi	35	1	1	2.9%	PASS
14637273	EA3AGB	3	1	1	33.3%	PASS
14909294	MAUSyagi	31	14	14	45.2%	PASS
14909617	W6MSU UHF	4	4	3	75.0%	PASS
14909703	MAUSyagi-AK	1	2	1	100.0%	PASS

RESULT

5/5 PASS across four stations | 22 valid local | 20 exact matches

Understanding Recovery Rate and Not Decoded

27% EXACT RECOVERY

- 20 exact matches divided by 74 controls.
- Measures local OGG reproduction under current settings.
- Affected by SNR, tuning, Doppler, gain, and clock recovery.

TWO DIFFERENT MEANINGS

- Validation: no exact local recovery or rejected diagnostic PDU.
- Dashboard: no authorized telemetry field mapping.
- Neither automatically means invalid spacecraft data.

RESULT

27% is audio recovery, not spacecraft health or total decoder accuracy.

KOYO BEACON DECODER

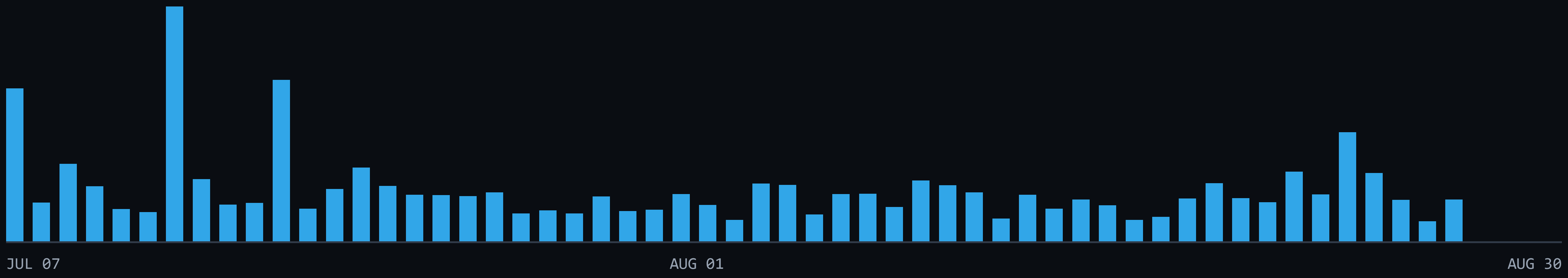
Historical Mission Coverage

55
UTC days

1,055
observations / 114 stations

17,155
SatNOGS-demodulated frames

2026-07-07 to 2026-08-30 UTC



RESULT

17,155 historical controls; direct byte-exact OGG proof covers five observations.

Stage 6 - Telemetry Confidence Contract

CONFIRMED

OBC time and counters
Battery TH0 / TH1
CDH and ADCS temperatures
PIB and SD health counters

CANDIDATE

Solar voltage channels 1 / 2
COMM raw voltage
Behavior plausible
Mapping or scale unverified

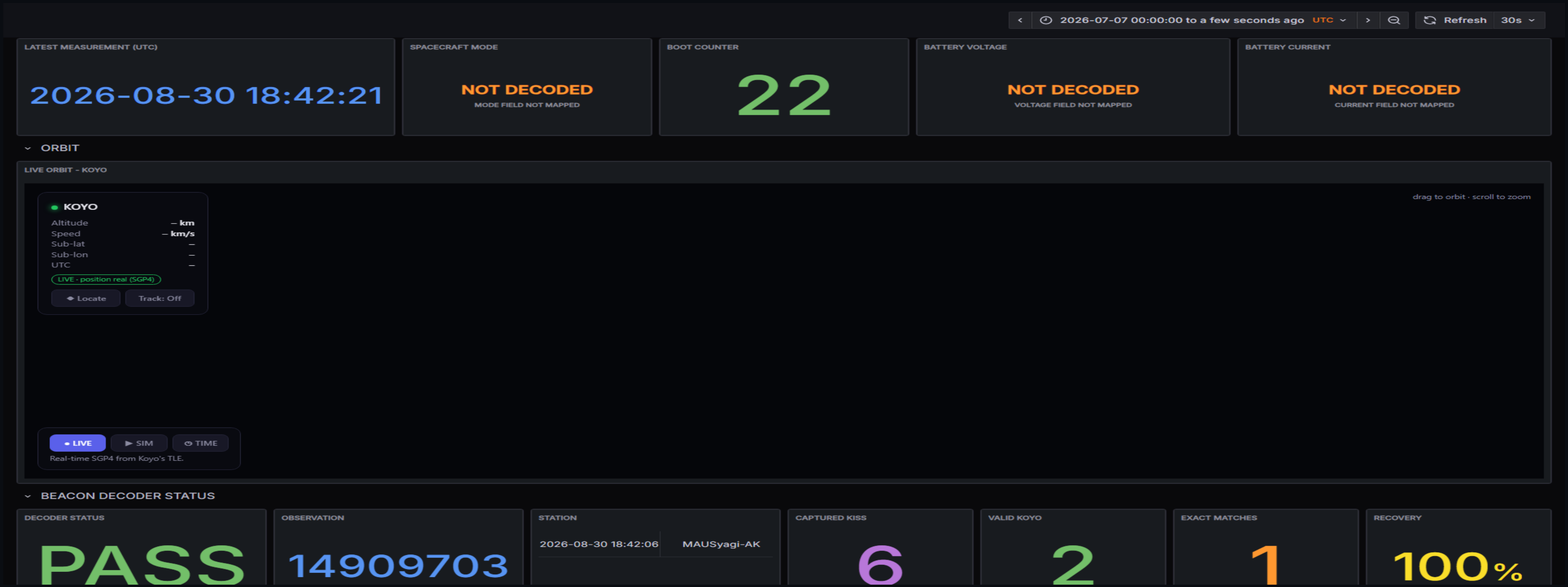
NOT DECODED

Spacecraft mode
Battery voltage / current
Solar currents
Power distribution fields

RESULT

Complete target structure with explicit confidence and zero fabricated values.

Stage 7 - InfluxDB and Full Grafana Dashboard



43 PANELS | decoder status + raw HEX | 22 / 22 Flux queries passed | no fabricated values

Reproducible End-to-End Demo

ONE COMMAND

- live_refresh.ps1 -ObsId 14909703
- Download OGG, create WAV, run GNU Radio.
- Filter, compare controls, and push the dashboard.

VERIFIED OUTPUT

- Six KISS frames; two valid KOYO frames.
- One byte-exact control match.
- InfluxDB HTTP 204; Grafana PASS plus raw HEX.

RESULT

SatNOGS audio -> local GNU Radio -> InfluxDB/Grafana completed successfully.

Limitations, Next Steps, and Conclusion

CURRENT LIMITATIONS

- 27% exact OGG recovery; five direct audio tests.
- SatNOGS is store-and-forward, not continuous RF.
- Candidate and unavailable telemetry mappings remain.

NEXT TECHNICAL STEPS

- Adaptive Doppler, tuning, and clock recovery.
- Stream OGG processing without permanent WAV files.
- Expand validation and authorize remaining mappings.

RESULT

The audio-to-frame path is proven; recovery optimization and mapping validation remain.